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The mechanical and oxidation properties of novel B2-ordered $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ refractory high-entropy alloys

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ABSTRACT

A series of novel B2-ordered $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ refractory high-entropy alloys (RHEAs) was designed and prepared by vacuum arc-melting. The effect of Al alloying on the microstructure, mechanical properties, and oxidation properties was investigated. The Al addition triggers the transformation of the crystal structure from disordered BCC phase to ordered B2 phase, and the Al1 alloy present single B2 phase structure. This B2 ordering results in significant yield strength increase of the RHEAs, i.e., from 915 MPa to 1410 MPa at room temperature (RT, leading to an excellent specific strength of ~232.3 KPa·m^3/kg for the Al1 alloy), 719 MPa to 1088 MPa at 873 K and 126 MPa to 278 MPa at 1073 K, respectively. However, in the expense, the near full and full B2-structured Al0.75 and Al1 alloys show dramatic reduction in compressive ductility down to 31.4% and 15.1% at RT. Four parameters, VEC, ΔH_{mix}, $\overline{Bo}$ and $\overline{Md}$ are employed to predict the B2 ordering in the RHEAs, and the calculation shows that ordered phases are preferred in the realm of $\Delta H_{mix} < -14$ kJ·mol^{-1} and $\overline{Bo} < 0.4\overline{Md}+1.9$. In addition, the introducing of Al also significantly enhances the oxidation resistance of the alloys. While the Al0 alloy shows catastrophic oxidation within 24 h in air at 1073 K, the Al1 alloy only gains 112 mg·cm^{-2} after 50 h. However, the constitution of the oxide layer remains loose and complex, consisting of multiple oxides including Al_2O_3, TiO_2, $(Zr,Hf)O_2$ and $(Zr,Hf)V_2O_7$.

1. Introduction

The development of high-entropy alloys (HEAs) or complex concentrated alloys (CCAs) challenges the traditional alloying strategy consisting of only one or two predominant elements [1,2]. The greatly broadened alloy composition space by the idea of HEAs results in unexpected solid-solution related structures and hence outstanding integrated properties, which attracts growing attentions in the material science community over the past decades [3–8]. Refractory high-entropy alloys (RHEAs) are an important group of HEAs that mainly consist of refractory elements. These elements render RHEAs body centered cubic (BCC) or hexagonal close-packed (HCP) structure with high strength and hardness, decent ductility and excellent corrosion resistance [9–11], thereinto, the high and soften resistance at elevated temperatures are the most potential advantage [12–14]. For example, WMoTaNb(V) can retain over 500 MPa at 1600 °C [15], yet their high density, room temperature (RT) brittleness, and poor oxidation resistance greatly limits the application potential of these W/Mo contained RHEAs [16,17]. In the consideration of a good balance of strength, ductility, density, and oxidation resistance, antioxidant elements with low density, e.g., Al, Cr, Si are usually added in RHEAs. These newly designed RHEAs include $AlMo_{0.5}NbTa_{0.5}TiZr$, AlTiVCr, $AlNbTiVZr_x$, $(Mo_{0.5}NbHf_{0.5}ZrTi)/M_5Si_3$ and $AlCr_xNbTiV$ [18–24], with densities in the range of 5.0 to 8.4 g/cm^3 and specific strengths superior to Ni-based superalloys, yet the ductility lose remains without decent improvement. This can be attributed to the formation of brittle intermetallic phases, i.e., B2 phase in the case of Al addition [25–27], Cr_2M Laves phase in the case of Cr addition [22,28,29], and M_5Si_3 silicide in the case of Si addition [21,30], respectively. While B2 phase keeps a coherent interface with the BCC matrix which totally differs from the other two intermetallic phases, the ductility loss of the Al-RHEAs can be the lowest, with compressive ductility >10%.

For oxidation performance of RHEAs, however, limited studies are carried out to date. Liu et al. investigated the oxidation properties of a series of Nb-Cr-Mo-Al-Ti/V/Si RHEAs with different content of Ti, V, and Si in early years [31]. They found that the oxidation resistance of the

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RHEAs was improved with Ti and Si addition, but deteriorated with V addition. Gorr et al. studied the oxidation performances of TaMoCrTiAl, WMoCrTiAl, NbMoCrTiAl(Si) RHEAs, all showing mass gain below 10 mg/cm^2 at 1000 °C and 1100 °C for 40 h and oxidation kinetics of parabolic pattern except NbMoCrTiAl [32–35]. Bulter et al. systematically investigated the oxidation properties of TiZrNbV/TiZrNbCr and $Al_{20}Nb_{30}Ta_{10}Ti_{30}Zr_{10}$ RHEAs [36,37]. TiZrNbV HEA presents a near-linear oxidation kinetic curve, while those of TiZrNbCr and $Al_{20}Nb_{30}Ta_{10}Ti_{30}Zr_{10}$ RHEAs are parabolic, with the mass gain of about 106 mg/cm^2 and 40 mg/cm^2 respectively, after 100 h of oxidation at 1000 °C. Due to the nature of compositional complex, these RHEAs form oxide layers with multiple complicated oxides that are different from traditional alloys, i.e., single, or a simple mix of the three classical oxides Al_2O_3, Cr_2O_3 and SiO_2. The jointly formation and complex distribution of these oxides in the oxide layers such as $NbAlO_4$, ZrO_2, $Ti_7Al_2O_{15}$, $Ta_2Nb_4O_{15}$. $TiNb_2O_7$, TiO_2, $Nb_2Zr_6O_{17}$ seems therefore responsible for the reported poor oxidation resistance. Although the heavy addition of Al and/or Cr would greatly improve the oxidation capacity of RHEA, their mechanical properties are inevitably sacrificed, in particular resulting in poor ductility at RT. Further studies focusing on the balance between the oxidation and mechanical properties are still urgently needed.

Our previous studies presented a successful design of a single BCC structure $Ti_2ZrHf_{0.5}VNb_{0.5}$ RHEA with a relatively low density of 6.67 g/cm^3 [38]. Moreover, it exhibits excellent RT mechanical properties, with compression yield strength of 1000 MPa and ductility over 50%. Based on this RHEA and following the idea of balancing oxidation and mechanical properties, in this work, we additionally introduced alloying of Al in the $Ti_2ZrHf_{0.5}VNb_{0.5}$ RHEA, in another words, designed a new series of $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ RHEAs, and systematically investigated the microstructure evolution and the corresponding mechanical responses at RT. In particular, we happened to find that when x increases to 1, the crystalline structure changes to full B2, an ordered intermetallic phase, which strongly influenced the mechanical behavior. The performances at elevated temperatures including oxidation and compression, were also involved, with the detailed corrosion and deformation mechanisms briefly assessed in the following.

2. Material and methods

Novel series of $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ RHEAs with various amounts of Al addition (simply denoted as Al0, Al0.25, Al0.5, Al0.75, Al1, respectively) were prepared by vacuum arc melting under a Ti-gettered argon atmosphere on a water-cooled copper crucible. Raw metals of Ti, Zr, V, Nb, Al with purity of 99.95 wt% and Hf with purity of 99.5 wt% were used. The ingots were flipped and remelted at least seven times to assure chemical homogeneity. All samples were cut by electric discharge machining, prepared by grinding, polishing in diamond suspension, and cleaned by ultrasound before further experiments.

The crystal structure was identified by X-ray diffraction (XRD, EMPYREAN) using Cu-Ka radiation target operated at 40 kV and 40 mA, and the angles of 2θ scanning was from 20° to 100° with a scanning rate of 4°/min. The microstructure characterization as well as the chemical composition was carried out by electron probe microanalyzer (EPMA, JXA-8530F PLUS) and scanning electron microscaopy (SEM, Zeiss Supra 55). More details of microstructure at nano-scale were revealed by transmission electron microscopy (TEM, Tecnai G2 F20), and the specimens for TEM were prepared by a dual-beam focused ion beam (FIB, Helios G4UX, Thermo Fisher Scientific USA) instrument.

Compression tests (test standards: ASTM E9–09) were conducted on cylindrical specimens with the dimension of Φ6 × 9 mm using a Wippermann (WPM) materials testing machine at room temperature and a dynamic thermal simulation testing machine (Gleeble-1500D) at high temperatures (873 K and 1073 K), respectively. The strain rate was set at 1×10^{-3}/s. During high temperature test, the specimens were first heated to the testing temperature at a heating rate of 5 K/s and held for 10 min before loading. Vickers hardness (test standards used ASTM: E384-11e1) was measured using a Vickers hardness tester (MH-50). The loading force is 1000 g and the loading time is 15 s. The measurements were repeated seven times on each sample. The density of the series alloys was measured by the Archimedes drainage method (Mettler-ME204E balance with precision of 0.0001 g).

Isothermal oxidation tests were conducted at 1073 K and exposed to static laboratory air. The dimensions of the prepared samples were 10 × 10 × ~3 mm after surface grinding down to 2000-grits. Mass gain of the oxidation samples was measured using a high sensitivity balance (0.0001 g) and recorded from 1 h to 50 h. Three samples for each time interruption were used to ensure data accuracy. The phase constitutions and microstructures of the oxidized samples were investigated by XRD and EPMA.

3. Results

3.1. Phase identification and microstructure

Fig. 1 shows the X-ray diffraction profiles of the as-cast $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ RHEAs. Only single disordered BCC phase peaks can be found in Al0, Al0.25 and Al0.5 alloys in Fig. 1(a), while in the RHEAs with higher Al content, i.e., Al0.75 and Al1 alloys, minor peaks belonging to the B2 crystal identified as the $(100)_{B2}$, $(111)_{B2}$ and $(210)_{B2}$ appear. This suggests that the Al addition triggers the B2-type ordering in the $Ti_2ZrHf_{0.5}VNb_{0.5}$ RHEA that show basically BCC crystalline structure. Fig. 1(b) shows the details of the XRD profiles from 25 to 40 degree. The pattern is smooth without any extra weak peaks beside those above from B2 and BCC. The lattice parameters were calculated decreasing from 3.3655 Å to 3.3269 Å as the Al content increases from x = 0 to x = 1.0. This could be roughly attributed to the smaller atom radius of Al (1.43 Å) compared to the average atomic radius (1.437 Å) of the $Ti_2ZrHf_{0.5}VNb_{0.5}$ matrix.

Fig. 2 presents the microstructures of the as-cast $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ RHEAs. Fig. 2(a)–(e) are the low magnification BSE images while Fig. 2 (f)–(j) are at higher magnification. Interestingly, typical dendritic morphology prevails in the alloy series regardless of the Al addition and in particular, the presence of B2-ordering. The only visual change lies on the size of the dendrite arm. With the increase of Al content, the primary dendrite becomes thinner, and the secondary dendrite increase and become more directional. The BSE contrast difference between dendrite (d-dark) and interdendrite (id-light) in these images also indicates the occurrence of the dendritic elemental redistribution during solidification, which is listed in the Table 1. We find that although the composition deviation between dendrite and interdendrite becomes stronger with the addition of Al, the intensity of this derivation is still quite weak, no larger than 4 at.%, agreeing well with the weak BSE contrast.

In order to clarify the possible B2/BCC phase transitions, we present TEM observations of Al0, Al0.25, and Al0.5 alloys firstly in Fig. 3, and those of Al0.75 and Al1 alloys in Fig. 4. Fig. 3(a) and (c) are the bright field (BF) images of Al0 and Al0.25 alloys, respectively. It can be seen that the contrast of the images is uniform. The corresponding selected area diffraction patterns (SADP) taken from red circled regions in Fig. 3 (b) and (d) confirm the single disordered BCC phase structure, with no other phase or B2 ordering exists. Fig. 3(e) is the BF images of Al0.5 alloys, similarly showing no significant contrast. However, the SADPs in two different regions along [001] and [011] zone axis, as shown in the insets in Fig. 3(f) and (g) together verify the existence of ordered B2 phase. The corresponding dark field images in Fig. 3(f) and (g) reveal clearly the uniform and dense distribution of the B2 phase particles with bright contrast, embedding in the BCC matrix. These B2 particles have a mean diameter of 2 nm and a volume fraction of 3.5%. The high-resolution (HR) images of the red rectangular region in Fig. 3(g) is shown in Fig. 3(h), which indicates the full coherency of the B2 phase and the BCC matrix.

Fig. 4(a)–(d) and (e)–(h) display the TEM results of the Al0.75 alloy

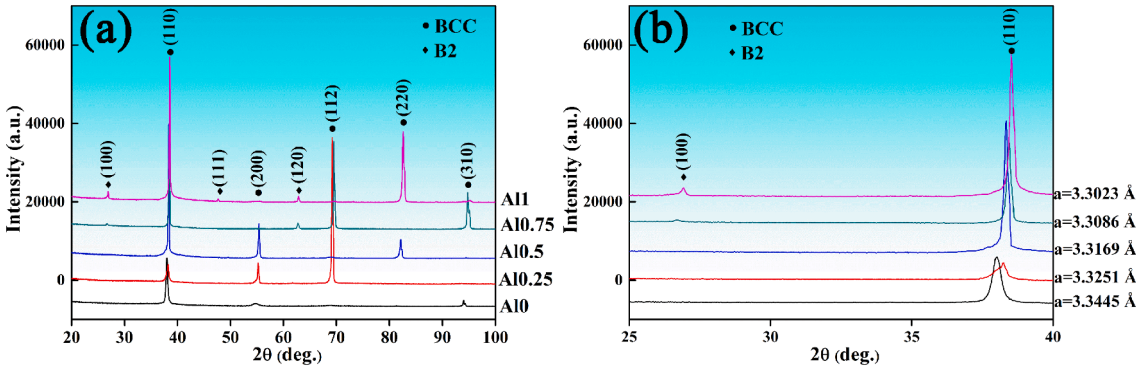

Fig. 1. (a) XRD patterns of $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ alloys, (b) enlarge view of (100) and (110) peaks.

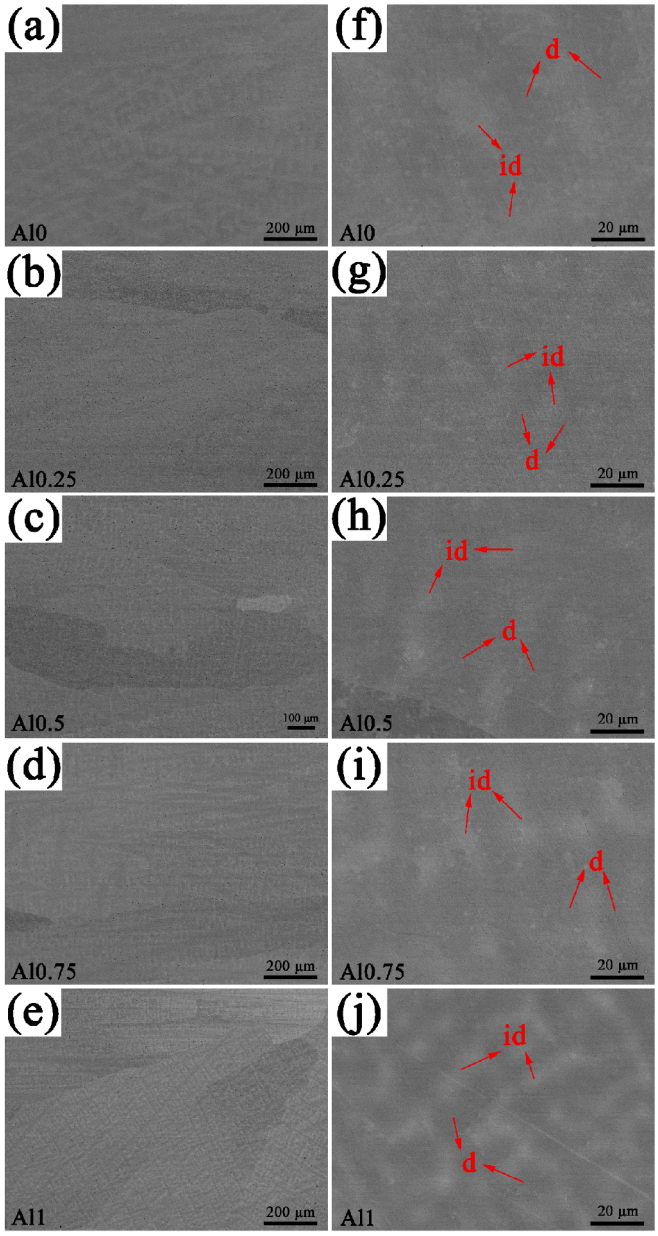

Fig. 2. Microstructures of of $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ alloys: (a)-(e) are the low magnified BSE images while (f)-(j) are the high magnification BSE images. (d and id represent the dendrite and interdendrite, respectively).

Table 1
Chemical composition (at.%) of dendrite (D) and interdendrite (ID) regions in $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ alloys.

Alloys	Regions	Ti	Zr	Hf	V	Nb	Al
Al0	D	37.86	21.79	11.17	16.84	12.34	0
	ID	35.31	22.20	11.42	19.71	11.36	0
Al0.25	D	36.49	19.58	10.53	17.28	11.74	4.38
	ID	35.98	20.61	10.31	17.76	10.58	4.75
Al0.5	D	35.73	18.54	9.79	16.73	11.26	7.95
	ID	35.04	19.59	9.97	16.65	10.40	8.35
Al0.75	D	33.85	18.06	9.63	16.30	10.44	11.72
	ID	31.22	20.66	10.09	15.65	10.09	12.29
Al1	D	32.95	17.16	9.22	15.50	9.93	15.24
	ID	30.93	20.34	9.37	15.46	8.48	15.41

and Al1 alloy, respectively. The bright field (BF) image in Fig. 4(a) reveals a dual-phase composite structure with different contrasts. The darker phase is particle-like and evenly embedded in the matrix of the lighter phase, with a volume fraction of 61.5%. Selected area diffraction pattern (SADP) along the [001] zone axis obtained from the red circle is shown in Fig. 4(b), where the typical B2 superlattice reflections are found. According to the two-dark field (DF) images taken at the disordered reflection spots, Fig. 4(c), and the ordered reflection spots, Fig. 4 (d), respectively, the darker phase in Fig. 4(a) can be identified with ordered B2 structure, while the lighter phase is of disordered BCC structure. We probed the composition of the two phases by EDS as also listed in Fig. 4(a), surprisingly, no significant compositional deviations exist. In order to more clearly reveal the differences in composition of the two regions, the TEM-EDS elemental mapping is also tested shown in Fig. 5, no difference in composition can be observed obviously. This supports the conjecture given above that the two phases form in an interbedded manner, in which rearrangement of the local atoms in the cubic lattice occurs and quickly develops into nanoscale domains without apparent long-range elemental diffusion. With the addition of Al element, the short-range ordering area of the alloys increases, and then the area of the B2 phase increases. Fig. 4(e) is the bright field (BF) image under the [001] zone axis of the Al1 alloy, where only plenty antiphase boundaries (APBs) appear with no other detectable phases or precipitations. This may indicate that the alloy is almost completely ordered. The SADP in Fig. 4(f) is almost the same as the B2 superlattice reflections given in Fig. 4(b). The high-resolution (HR) images of the rectangular region in Fig. 4(e) is shown in Figs. 4(g), and (h) is the magnified HR-image transformed by Fast Fourier Transform (FFT) and inverse Fast Fourier Transform (IFFT) of the square region as marked in Fig. 4(g). The inserted FFT diffraction pattern again evidences the strong B2 reflection. Similar APB morphology with B2 superlattice reflection can be found in other pure B2 alloys including TiAl, NiAl and also the earlier report of Ta/MoCrTiAl and TiVNbCr RHEAs, suggesting that here the Al1 alloy is of a pure B2 structure [19,39,40,41–43].

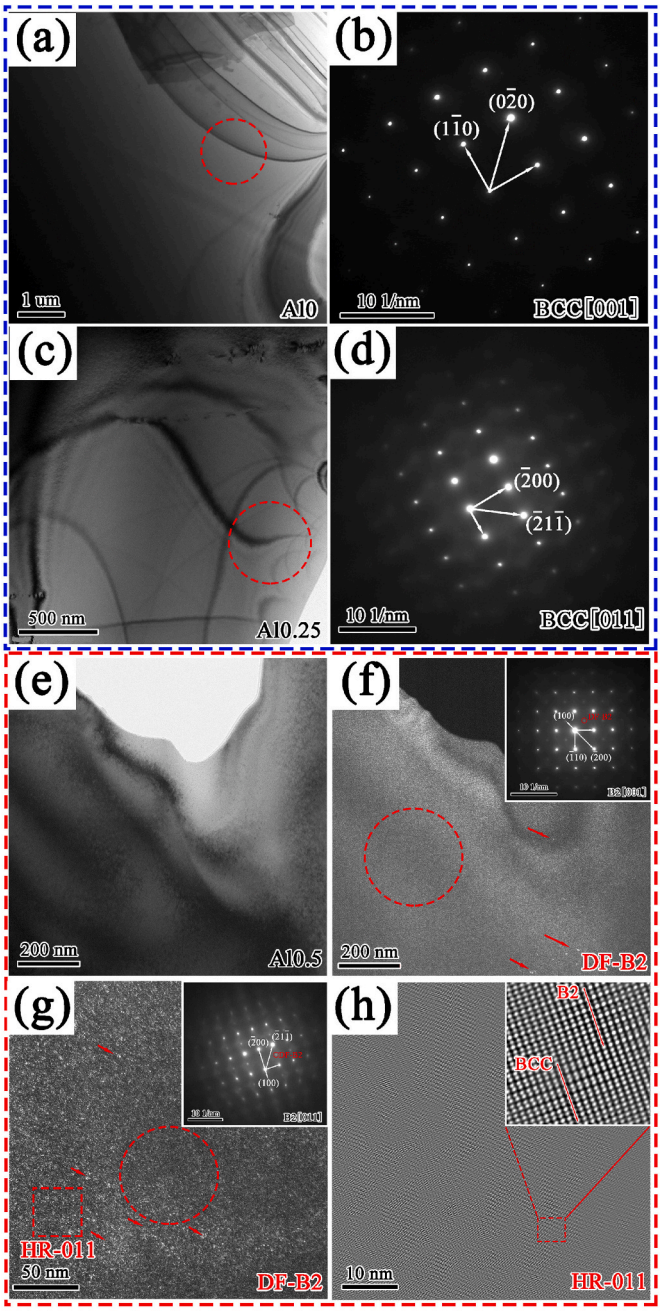

Fig. 3. (a), (c) BF microstructure image of as-cast Al0 and Al0.25 alloys, respectively; (b) the SADPs along the [001] zone axis in red circle of Al0 alloy; (d) the SADPs along the [011] zone axis in red circle of Al0.25 alloy; (e) BF microstructure image of as-cast Al0.5 alloy; (f)-(g) the DF images encased in ordered spot along the [001] and [011] zone axis; (f) HR-TEM images of the red square position in (e) transformed by FFT and IFFT (the upper right corner of picture is the enlarge images of HR-TEM).

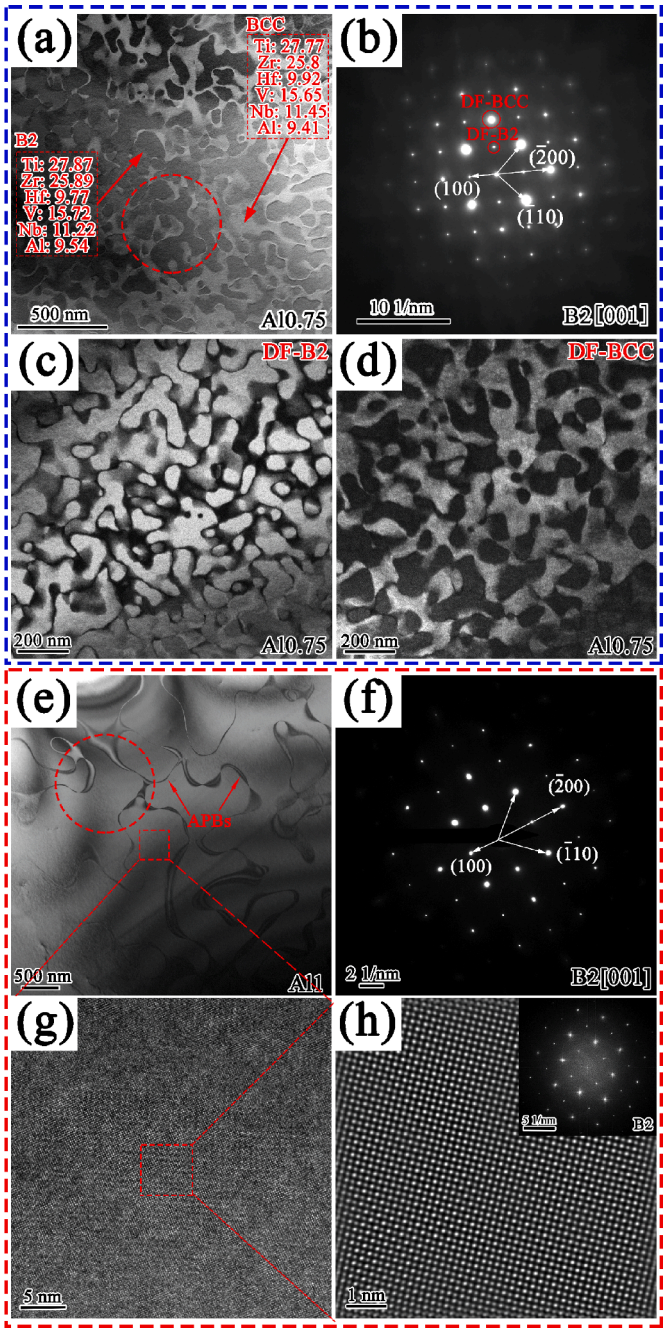

Fig. 4. (a), (e) BF microstructure image of as-cast Al0.75 and Al1 alloys, respectively; (b), (f) the SADPs along the [001] zone axis in red circle of Al0.75 and Al1 alloys, respectively; (c)-(d) the DF images encased in disordered spot and ordered spot; (g) HR-TEM images of the red square position in (e); (f) HR-TEM images transformed by FFT and IFFT (the upper right corner of picture is the corresponding FFT-SADPs).

3.2. Mechanical properties

The RT compressive engineering stress-strain curves are depicted in Fig. 6(a), and the extracted yield strength (σ_y) and hardness values as function a of the Al addition, i.e., x, in the present RHEAs are given in Table 2 and described in Fig. 6(b). As x increases from 0 to 1, σ_y exhibit a significant increase, from 915 MPa to 1410 MPa, i.e., around 54%. The hardness of the RHEAs shows similar linear increment as a function of x, as evidenced in Fig. 6(b). Moreover, if the unit of hardness is converted to MPa, it can be found that the hardness of the series alloys approximately triples their yield strength. And the average ratio of hardness to strength is $\varphi = HV/\sigma_y = 3.16$. it is close to the value of $\varphi \approx 3$, which is the common proportional relationship for pure metals and commercial alloys [44]. The compressive fracture strain (ε_f) remains over 50% when $x \leq 0.5$, however, it dramatically decreases to 31.5% and 15.1% in the Al0.75 and Al1 alloys, respectively. In addition, the series of Al-modified RHEAs show much higher specific yield strength (SYS) when compared with other reported RHEAs due to the sharp decrease of density from 6.68 g/cm^3 to 6.07 g/cm^3 as the addition of light-weight element Al. Detailed analysis can be found in the following discussion section.

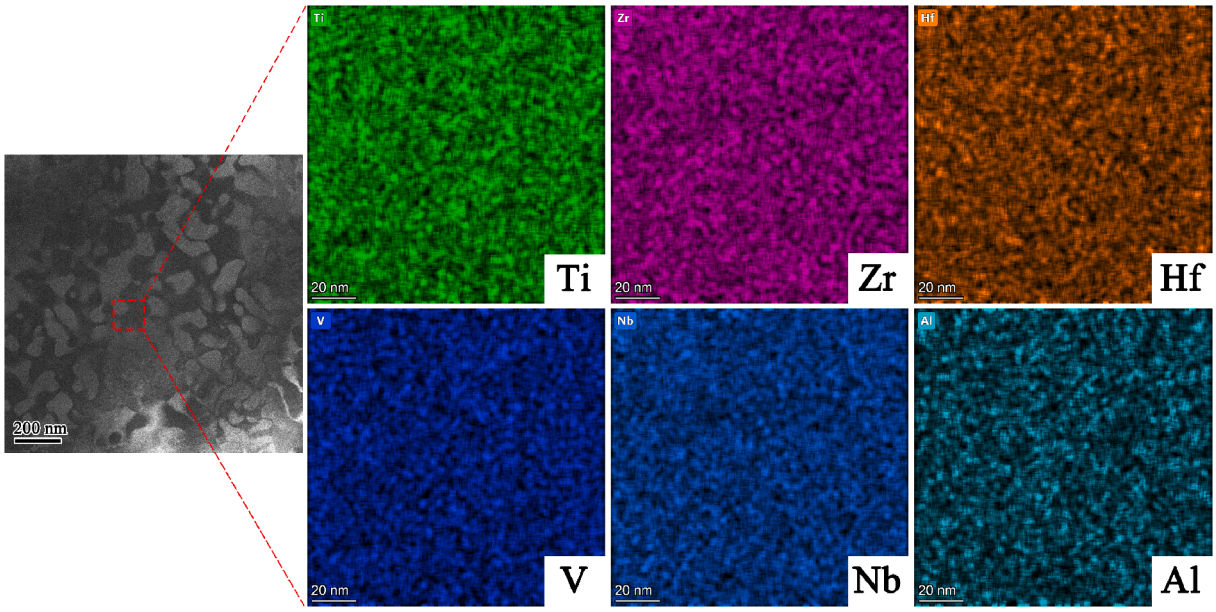

Fig. 5. TEM-EDS elemental mappings of Ti, Zr. Hf, V, Nb, Al for Al0.75 alloy.

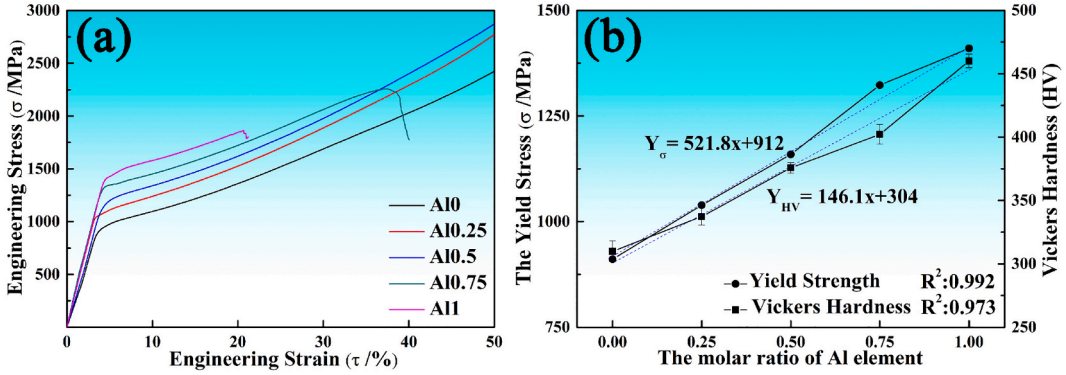

Fig. 6. (a) the compressive engineering stress-strain curves of $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ alloys at RT; (b) linear graphs of Vickers hardness and yield strength for $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ alloys.

Table 2
Mechanical properties of the $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ alloys: yield strength (σ_y/MPa), plastic strain (ε_f/%), fracture strength (σ_b/MPa), Vickers hardness (HV/MPa), density (ρ/g·cm^{-3}) and specific yield strength (SYS/KPa·m^3·kg^{-1}).

Alloys	σ_y	ε_f	σ_b	HV/MPa	ρ	SYS
Al0	915	>50	–	309.8/3036	6.68	136.9
Al0.25	1040	>50	–	337.4/3306	6.51	159.8
Al0.5	1159	>50	–	375.7/3682	6.35	182.5
Al0.75	1323	31.4	2256	402.2/3942	6.21	213.1
Al1	1410	15.1	1863	460.0/4508	6.07	232.3

The high temperature compression engineering stress-strain curves of Al0, Al0.5 and Al1 alloys at 873 K and 1073 K are shown in Fig. 7(a) and (b), with the extracted parameters including σ_y, ε_f and SYS listed in Table 3. Similar to the case at RT, the addition of Al in the RHEA prominently rises σ_y at both temperatures, from 710 MPa to 1088 MPa at 873 K and from 126 MPa to 278 MPa at 1073 K, respectively. Noticing that the Al1 alloy can maintain a yield strength over 1 GPa at 873 K, higher than many other RHEAs [21,53,67–70]. Fig. 7(c) shows the histogram of the yield strength change at different temperatures. The high strength of the three alloys indicates that the alloys exhibit resistance to softening up to 873 K.

3.3. Oxidation properties

The isothermal oxidation kinetic curve of a metallic material can be usually described in the form of Power law [74]:

$$\Delta m = kt^n \quad (1)$$

where Δm is mass gain per unit surface area (mg/cm^2), t is the holding time (h), k is the oxide growth rate constant, and n is the time exponent. This Eq. (1) can be transformed to a linear relationship:

$$lg(\Delta m) = lgk + nlgt \quad (2)$$

where the slope is the time exponent (n).

Fig. 8(a) show the fitted oxidation kinetics curves of the Al0, Al0.5 and Al1 alloys, while Fig. 7(b) presents the above curves in a Log/Log scale. The fitted n and k values are listed in the Table 4. The alloy without Al (Al0 alloy) clearly exhibit a linear oxidation behavior (n = 1.04) with k = 11.65 mg/(cm^2·h), suggesting that no oxide film formed on the Al0 alloy. As a consequence, the Al0 alloy suffered catastrophic oxidation after 24 h exposure. In contrast, the oxidation mass gain (Δm) of both the Al0.5 and Al1 alloys was notably decreased, and the oxidation kinetics can be divided into two parts according to the Log/Log curves in Fig. 8(b). Before 24 h, both the two curves obey the parabolic

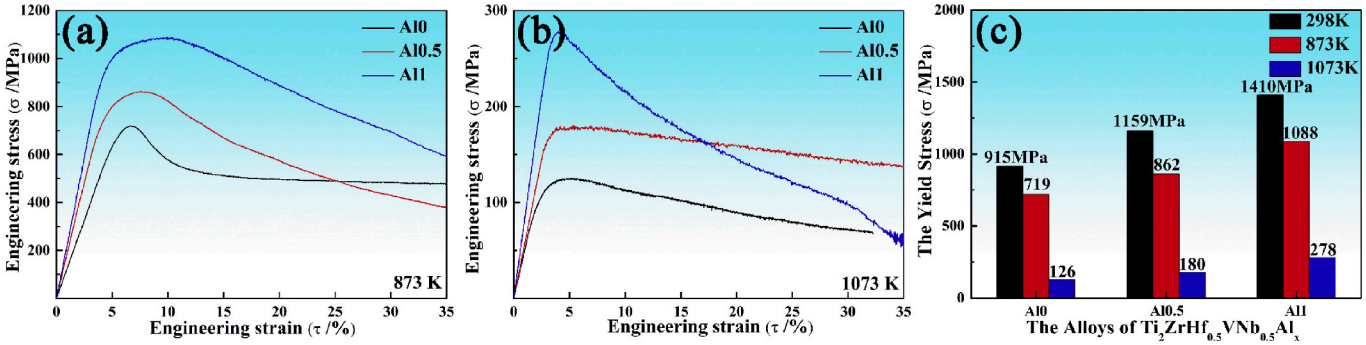

Fig. 7. Compressive engineering stress-strain curves of the Al0, Al0.5 and Al1 alloys at (a) 873 K and (b) at 1073 K. (c) Columnar graphs of yield strength at different test temperature.

Table 3
Mechanical properties of the $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ alloys at RT, 873 K and 1073 K: yield strength (σ_y/MPa), plastic strain (ε_f/%), and specific yield strength (SYS/KPa·m^3·kg^{-1}).

Alloy	Temperature (K)	σ_y	ε_f	SYS
Al0	RT	915	>50	136.9
	873	719	>35	107.6
	1073	126	>35	18.9
Al0.5	RT	1159	>50	182.5
	873	862	>35	135.7
	1073	180	>35	28.3
Al1	RT	1410	15.1	232.3
	873	1088	>35	179.2
	1073	278	>35	45.8

law, and the fitted n values of Al 0.5 and Al1 alloys is 0.62 and 0.67, respectively, indicating of possible oxygen protection mechanism functioned on the Al0.5 and Al1 alloys before 24 h. However, after 24 h, the oxidation kinetic curves changed towards linear relationship, indicating the loss of the protection ever since.

Fig. 9(a) and (b) present the macroscopic morphology as well as the cross-sectional microstructure evolution during oxidation. Oxidation samples, with different Al contents (x = 0, 0.5, 1) and different heat exposure time (10 h, 24 h, and 50 h) are selected. Notice the Al0 alloy is oxidized throughoutly after 24 h. For all the three RHEAs after oxidation over 10 h, cracking and warping occurred in the oxide layer from the bulk edges. The increase in Al content greatly suppressed such phenomenon, leaving the oxide layers much more completed in the Al1 alloy even after 50 h oxidation. From Fig. 9(b), we found that the Al0 alloy shows an oxidation depth of over 590 μm after only 10 h oxidation, while that of Al0.5 and Al1 alloys are much lesser, i.e., 384 μm and 363 μm, respectively. The thickness of the oxide film on the Al0.5 alloy increases to 900 μm and 1475 μm for 24 h and 50 h, while that of the Al1 alloy increases to 718 μm and 1350 μm for 24 h and 50 h, suggesting an overall better oxidation resistance of the Al1 alloy compared to the Al0.5 alloy. In support of this, we observe multiple cracks in the oxide layers of Al0.5 alloy after 10 and 24 h oxidation, and these cracks grow and finally join into larger cracks when further oxidized for 50 h. On the contrary, no cracks exist in the Al1 alloy after 10 and 24 h oxidation while the cracks after 50 h oxidation are obviously smaller. This finding seems agree with the oxidation kinetic results in Fig. 8(b), and suggests that the sudden change of the oxidation response is directly related to the formation of the cracks in the oxidation film that breaks down the protection layer and leads to the catastrophic oxidation.

4. Discussion

4.1. Effect of Al addition on the phase evolution

The formation of ordered B2 phase is of particular interest due to the near-perfect coherency relationship with the major BCC crystalline

Table 4
Fitted oxidation constants (k) and time exponent (n) with regression coefficient (R^2) calculated from the oxidation kinetics curves at 873 K and 1073 K of the Al0, Al0.5, Al1 alloys.

Alloys	Time	R^2	k	n
Al0	Before 24 h	0.99	11.65	1.04
Al0.5	Before 24 h	0.99	8.14	0.62
	After 24 h	0.99	1.70	1.12
Al1	Before 24 h	0.99	5.36	0.67
	After 24 h	0.98	1.52	1.10

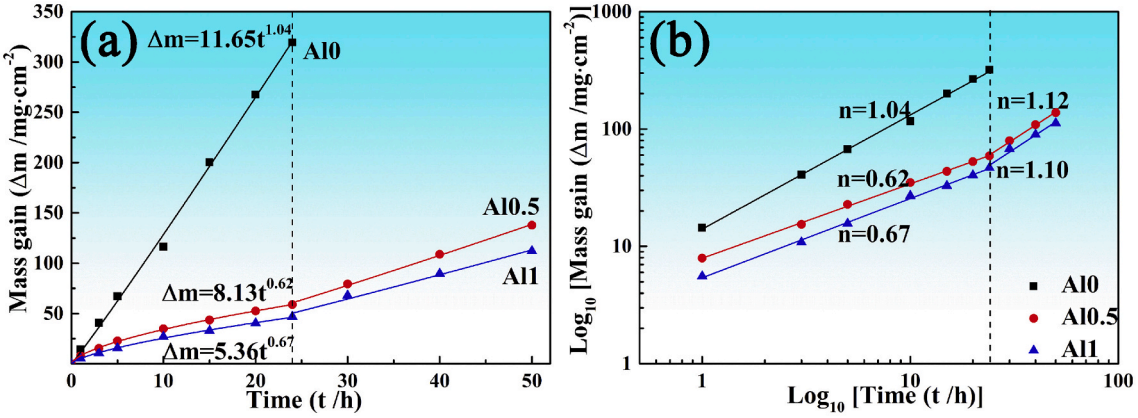

Fig. 8. Specific mass change versus time for the Al0, Al0.5 and Al1 alloys at 1073 K (a) on a linear scale; (b) and on the Log/Log scale.

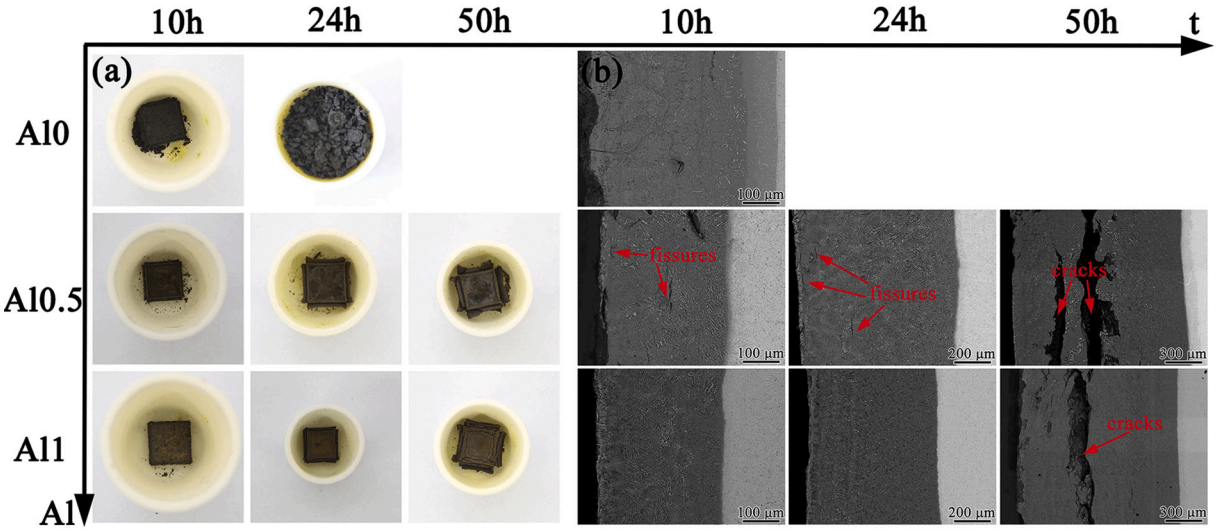

Fig. 9. (a) The macro morphology of the Al0, Al0.5 and Al1 alloys being oxidized after different times (10 h, 24 h and 50 h); (b) the cross-BSE images of Al0, Al0.5 and Al1 alloys after oxidation.

structure in RHEAs. Actually to date, most of the reported RHEAs containing, or composed of, B2 phase are alloyed with Al. This is because of the strong directional bonding between atoms of Al, and Ti/Zr, two principal elements in RHEAs, which leads to the B2-type TiAl or ZrAl crystalline structure. In the current RHEA series, it is reasonably to speculate that, the more Al addition, the higher tendency for the B2 phase to form. However, the real questions are, can we precisely predict that at which Al level the B2 phase starts to appear, and how/where the B2 ordering takes place.

Here for the very first question, we applied several frequently-used phase formation criteria including the valence electron concentration (VEC), difference of atomic sizes (δ), mixing enthalpy (ΔH_{mix}), mixing entropy (ΔS_{mix}), and Ω - parameter. And the definitions of these criteria are as following [42–44]:

$$VEC = \sum c_i (VEC)_i \tag{3}$$

$$\delta = \sqrt{\sum_{i=1}^{N} c_i \left(1 - r_i / \sum_{i=1}^{N} c_i r_i\right)^2} \tag{4}$$

$$\Delta H_{mix} = \sum_{i=1, j\neq 1}^{N} 4\Delta H_{ij}^{mix} c_i c_j \tag{5}$$

$$\Delta S_{mix} = -R \sum_{i=1}^{n} c_i ln c_i \tag{6}$$

$$\Omega = \frac{T_m \Delta S_{mix}}{|\Delta H_{mix}|} \tag{7}$$

where c_i and r_i are the atomic percentage and the Goldschmidt atomic radius of the *ith* element, respectively [41]. $(VEC)_i$ is the valence electron concentration of element *i*; H_{ij}^{mix} is an enthalpy of mixing between elements *i* and *j*; ΔS_{mix} is the mixing entropy of alloy; T_m is the melting point calculated using the rule of mixture.

In general, when in the ranges of -15 kJ/mol $< \Delta H_{mix} < 5$ kJ/mol, $\delta < 6.6\%$ and $\Omega \geq 1.1$ [42,43], HEAs tend to present as a solid solution. Also, BCC phase likely shows in HEAs at VEC < 6.87 while FCC phase forms at VEC > 8 [44]. We calculate and list all above parameters of the current RHEAs in Table 5, and plot VEC and ΔH_{mix} values in Fig. 9(a), with reference from other reported BCC HEAs. The VEC values of this RHEA series are all less than 4.3, and decrease with the increase of Al content. This indicates that the Al-RHEAs all prefer a BCC structure. However, Al slight decreases the atomic radius difference, and increases the absolute system enthalpy of the alloys hence decreases Ω, suggesting that beside the BCC main phase, other intermetallic compounds (here is the B2 phase), may form due to the addition of Al, i.e., when $\Delta H_{mix} < -14$ kJ/mol. According to Fig. 9(a) this criterion is in good accordance with the referred BCC HEAs.

We also involve another criterion, i.e., Bo-Md in this study. This criterion is proved effective in phase prediction yet in traditional Ti alloys [45], and most recently in the $TiZrHfNb_x$ RHEAs with a decent accuracy [46]. The value of Bo represents the bonding strength of Ti and other alloying elements, and that of Md is mainly related to the electronegativity and metallic radius of elements. The system values of $\overline{Bo}$ and $\overline{Md}$ are calculated by rule of mixtures [45]:

$$\overline{Bo} = \sum c_i Bo_i \tag{8}$$

$$\overline{Md} = \sum c_i Md_i \tag{9}$$

where c_i is the atomic percentage of the *ith* element, Bo_i and Md_i are the values of Bo and Md parameters of element *i*, respectively.

We then plot $\overline{Bo}$ and $\overline{Md}$ values of the current RHEAs in Fig. 10(b), again with reference from the same BCC HEAs in Fig. 9(a). We can roughly divide the data in the figure into two parts by a line of $\overline{Bo}$=0.4$\overline{Md}$+1.9 as marked. RHEAs with the BCC solid solution structure lie in the top left of the graph, while most RHEAs with B2 phase are located in the bottom right. It means that lower $\overline{Bo}$ and higher $\overline{Md}$ promote the formation of B2 phase in RHEAs, which seems consistent with the formation rule in Ti alloys.

4.2. The effect of Al elements on the mechanical properties

As mentioned in the previous results, the addition of Al element significantly increases the strength of the alloys both at RT and high temperatures, regardless the slight decrease in atomic radius difference (δ) (from 5.81% to 5.31%) hence the possible decease of solid-solution hardening. With the addition of Al element, the width of the primary and secondary dendritic arms is all decreased, leading to the reduced dislocation free path. This will certainly yield additional strengthening effect according to the Hall-Petch relationship: $\sigma_y = \sigma_0 + kd^{-0.5}$. where σ_y is the yield strength, d is grain size and here can be estimated by the average width of the dendritic arms, σ_0 is a constant that can be considered as the lattice friction resistance, k is a constant related to the

Table 5
Parameters of VEC, δ, ΔH_{mix}, ΔS_{mix}, Ω, Bo, Md and T_m for the $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ alloys.

Alloy	VEC	δ (%)	ΔH_{mix} (kJ/mol)	ΔS_{mix} (J/K mol)	Ω	Bo	Md (eV)	T_m (K)
Al0	4.3	5.81	−0.72	12.23	36.62	2.915	2.480	2156
Al0.25	4.24	5.67	−6.04	13.24	4.60	2.892	2.467	2098
Al0.5	4.18	5.54	−10.41	13.65	2.68	2.871	2.454	2045
Al0.75	4.13	5.42	−14.02	13.85	1.97	2.851	2.443	1996
Al1	4.08	5.31	−17	13.94	1.6	2.834	2.433	1952

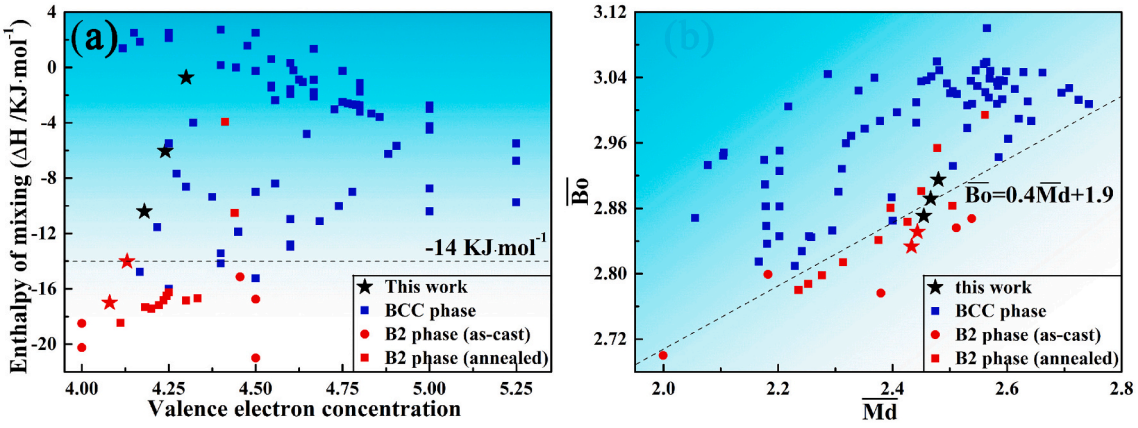

Fig. 10. (a) ΔH_{mix} - VEC and (b) $\overline{Bo}$ - $\overline{Md}$ scatter diagram of the $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ alloys and RHEAs from ref. [14,19,20,27,24,39,46,47]–[50–55,56,57,58–65,66–68].

material of the alloys and can be referred as 250 MPa·μm^{-1} according to [46,50,51.48,49]. The strength increment caused by the dendritic refinement can be estimated as 29 MPa, which is quite low regarding the total yield strength. Therefore, the high yield strength of the current RHEAs can be mainly attributed to the occurrence of the phase transformation i.e., the B2 ordering. Ordered crystalline structure is generally known stronger than their solid-solution counterparts due to the higher crystalline resistance to dislocation movement, including slip at RT case and climb at high temperature case. Extra driving force is required, and also extra energy has to be consumed, as the formation of a specific kind of fault, APB, when a dislocation shear across the crystal lattice and destroy the perfectly ordered atomic stacking. The APB energy is a parameter representing the difficulty for APB formation in the ordered crystal and therefore indexes the shear resistance for dislocation.

SYS is also important to evaluate the mechanical properties of structural alloys with particular consideration on the light weight of materials. Here we summarized and compared SYS of the current RHEAs series and many different Al/Cr contained RHEAs at RT. The scatter diagram of SYS and density at RT is shown in Fig. 11(a), where the points colored in black indicate the alloys under the as-cast condition, while the points in yellow show the alloys having hot isostatic pressure (HIP) or annealing treatment. Red stars represent the result in this work. It is clear that the current $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ alloys have remarkable high SYS with decent ductility. In addition, more than half of the points are on the left side, indicating of a poor compressive plasticity (<15%). Generally, excessive addition of Al and Cr in the RHEAs will trigger the vast precipitation of intermetallic compounds, which would greatly improve strength but sacrifice most of ductility. For example, the Cr_2M Laves phase forms in the Ti-contained RHEAs by the addition of Cr. However, in this study no complex compounds formed by the addition of Al to the $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ alloy except the B2 ordering. Ordered BCC phase has more slip systems than the complex Laves phase and is coherent with the BCC matrix. As a result, the near full B2 and B2 structured Al0.75 and Al1 alloy show comparable ductility at RT and thus a SYS of 232 KPa·m^3/kg for the Al1 alloy. This suggests that the addition of Al in the RHEAs to trigger coherent ordered phase transformation is a promising design strategy to improve the mechanical response.

We further plot the scatter diagram of SYS and density at 873 K in Fig. 11(b). It is observed that most classical RHEAs without Al show relatively low SYS at this temperature, while those contain Al exhibit relatively high SYS but poor ductility (Mostly less than 10%). We also see that the CoFeNiAl-contained HEAs show either good plasticity or decent SYS. For the three investigated RHEAs in this work, the base Al0 alloy have low specific strength, while after Al addition, their SYSs are greatly improved with ductility larger than 35%. In particular, the SYS of the Al1 alloy can reach to 180 kPa·m^3/kg, outperforming most reported RHEAs at 873 K.

4.3. The effect of Al elements on the oxidation properties

Fig. 12 shows the XRD patterns of the oxide layers of the Al0, Al0.5 and Al1 alloys after oxidation with different holding times at 1073 K. It indicates that all the three alloys form multiple complex compositional oxides instead of simple single oxide when suffered heat exposure. The oxide layer of Al0 alloy after 10 h oxidation is consisted of TiO_2, (Zr,Hf)O_2, (Ti,Zr)Nb_2O_7. No V-oxides can be detected, probably due to the formation of V_2O_5 oxide with low melting point that evaporates at 650 °C or above [73]. Together with the observation in Fig. 8, it can be roughly concluded that the reason for the poor oxidation resistance of the Al0 alloy without Al is: (a) no dense and protective oxide layer formed after oxidation, the oxidation layer of Al0 alloy cannot provide adequate resistance to the ingress of oxygen and was dispersed all throughout the oxide scale; and (b) the escape of the V_2O_5 oxides from the sample surface that leaves detrimental pores or holes, further accelerating the oxidation process [36]. With the addition of Al, i.e., in the Al0.5 and Al1 alloy, the new diffraction peaks of Al_2O_3 and (Zr,Hf)V_2O_7 appears in the XRD patterns after 10 h oxidation as shown in Fig. 11(b), however disappeared again after prolonged oxidation at 24 h and 50 h, as shown in Fig. 11(c). To reveal the detailed elemental information during the formation of the oxide layer in the two alloys, we further plot the elemental mapping of the Al1 sample after 10 h/24 h oxidation in the longitudinal section in Fig. 13. It can be found that the protective oxide Al_2O_3 is densely distributed on the top of oxide layers

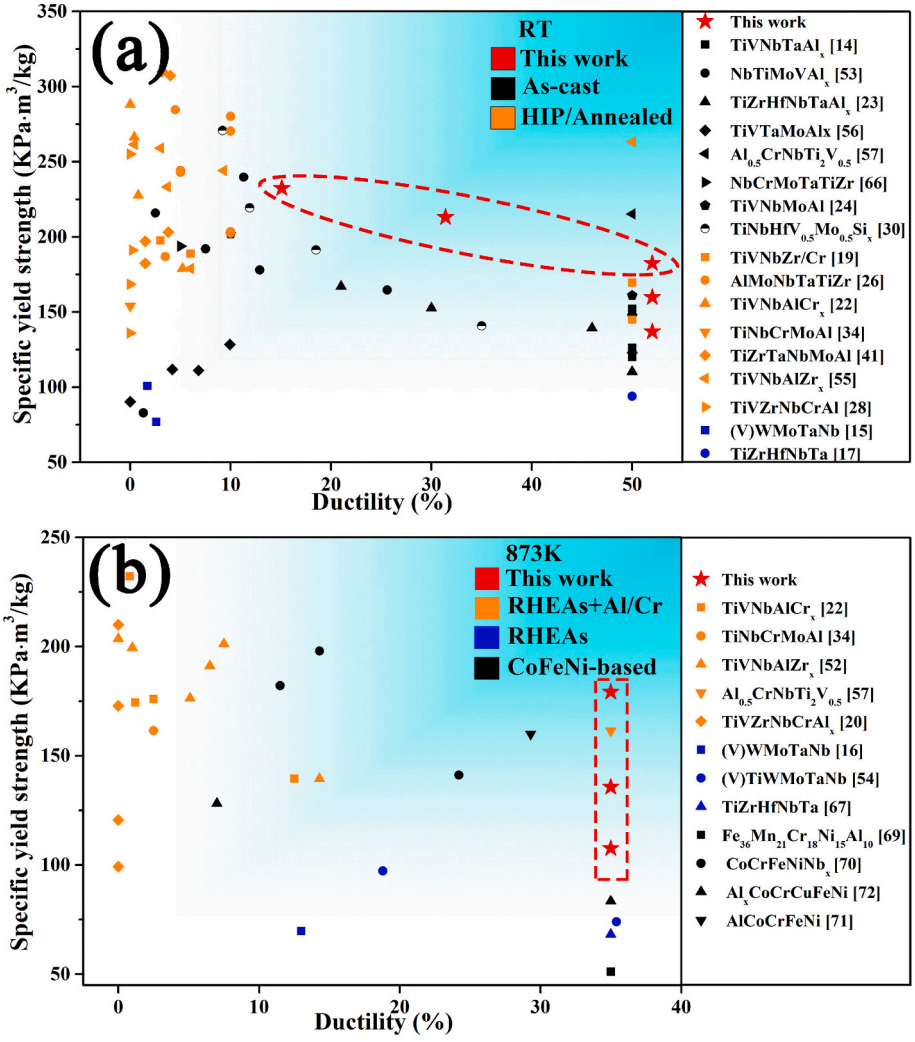

Fig. 11. SYS and ductility of the $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ alloys (a) at RT and (b) 873 K compared with other reported HEAs [14,15–20,22,24,26,28,30,34,41,52–55,56,57,66–72].

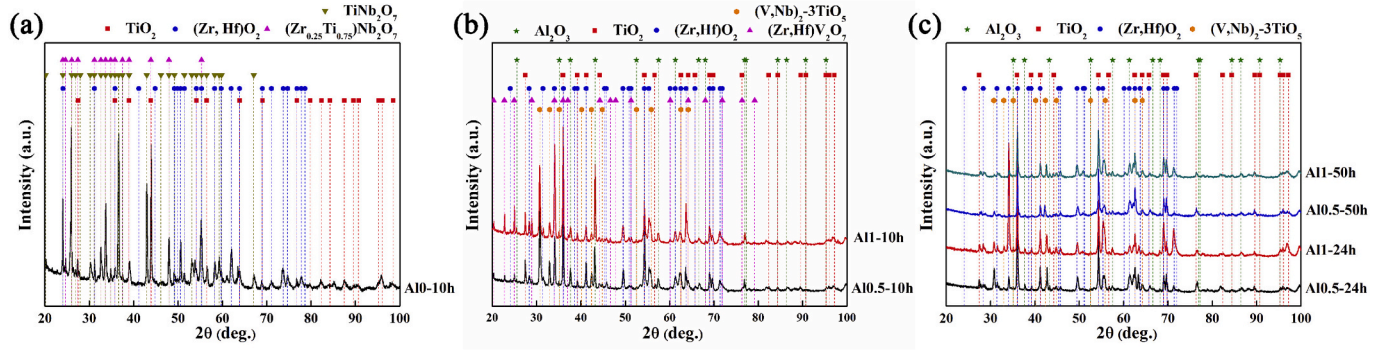

Fig. 12. XRD patterns of the oxide layers of the alloys that have been oxidized at different times ((a) Al0-10 h; (b)Al0.5/Al1-10 h; (c)Al0.5/1-24 h/50 h).

on the Al1 alloys with a thickness of ~2 μm at 10 h, and increase to 3–4 μm at 24 h. The Al_2O_3 layer is quite steady and stable at 10 h, which is able to protect the alloys from fast oxidation at earlier stage of heat exposure. Due to this outermost Al_2O_3 oxide layer, V_2O_5 oxide does not evaporate directly into the outside. Instead, it reacts with ZrO_2 and HfO_2 to form $(Zr,Hf)V_2O_7$ in the reaction of: $ZrO_2/HfO_2 + V_2O_5 = (Zr,Hf)V_2O_7$. This reaction suppresses the formation of volatile oxides V_2O_5, and become much stable when compared with V_2O_5, further slowing down the oxidation process in the Al0.5 and Al1 alloys. However, when at 24 h, the layer starts peeling and twisting, a symption of layer breakdown. This phenomenon can be explained by the Pilling-Bedworth (PBR) mechanism, where the formation and quick growth of multiple complex oxides inevitably expands and deforms the oxide layer, leading to the final failure. After 24 h and 50 h oxidation the disappearance of the diffraction peaks of $(Zr,Hf)V_2O_7$ likely suggests that the V_2O_5 oxide may generate again and thereby break through the protection layer. All

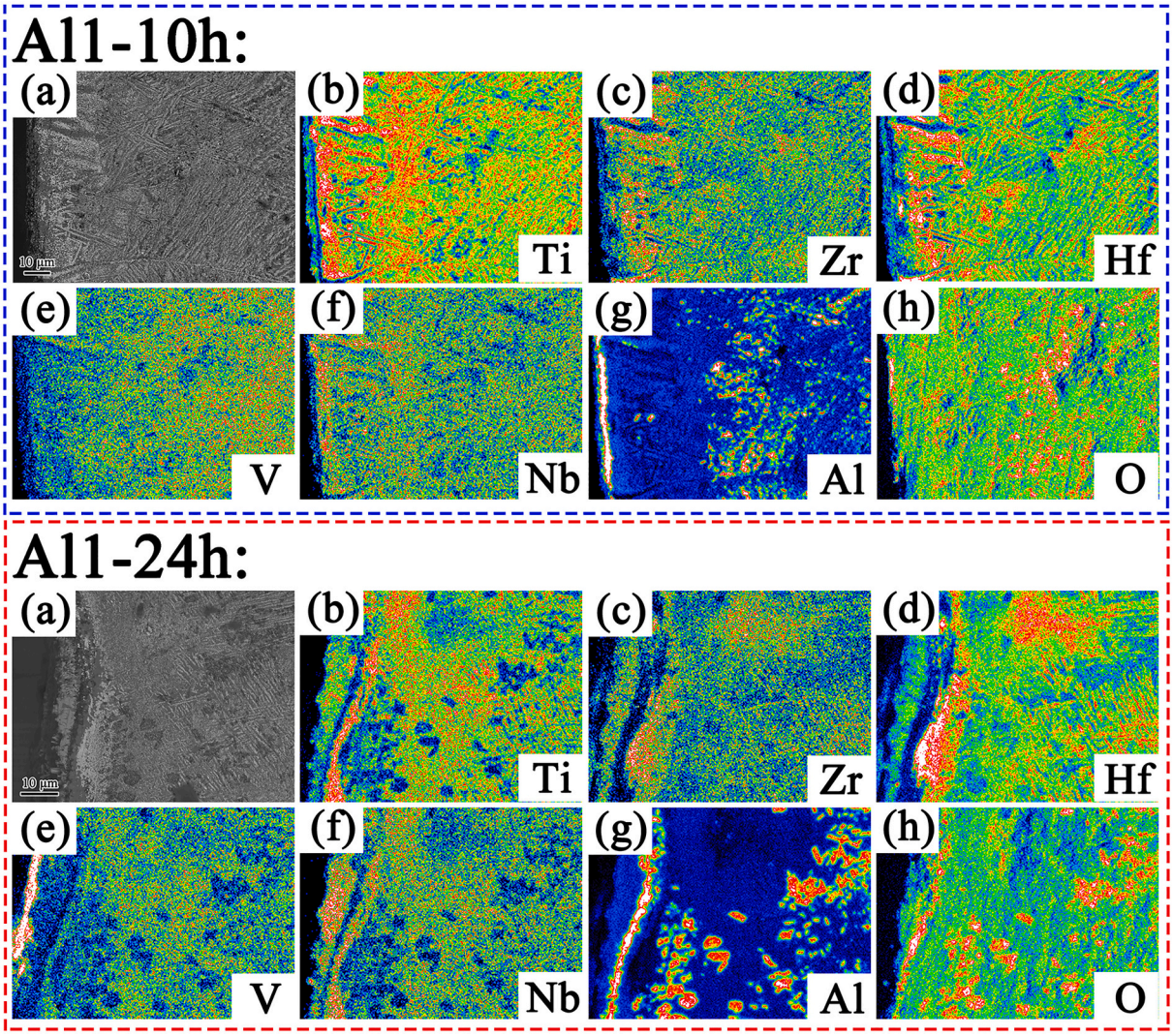

Fig. 13. Elemental mapping in the vertical section of the oxide layer after 10 h and 24 h of the Al1 alloys.

these phenomena indicate that the Al0.5/Al1 alloys oxide layers cannot effectively prevent oxidation after prolonged exposure particularly 24 h. This observation is consistent with the oxidation kinetics curves in Section 3.3, where oxidation acceleration begins at 24 h. In summary, the addition of Al element significantly enhances the oxidation resistance of the series alloys and reduces the oxidation weight gain after oxidation, but the effect is so limited and will soon disappear after oxidation with time larger than 24 h. V element is one of the crucial elements responsible for the poor oxidation performance. Further studies are needed to improve the oxidation properties of V-containing RHEAs via building long-term dense and stable protection layer containing multiple complex oxides.

5. Conclusion

In this work, the antioxidant light-weight element Al was added into the $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ RHEAs to improve the mechanical and oxidation properties, with main conclusions can be drawn as following:

1. Al plays a key role in the B2 ordering in the series of RHEAs. The addition of Al triggers the phase transformation in the $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ RHEAs from disordered BCC phase to ordered B2 phase without apparent long-range elemental diffusion. With the increase of Al content, the ordered B2 phase appears in the Al0.5 alloy in a very small range of short-range order, then expand to a large range of long-range order, and finally a single-phase B2 structure when x = 1. Three semi-empirical criteria are used to predict this phase formation, and we found that the B2 ordering starts when $\Delta H_{mix} < -14$ kJ·mol^{-1}, $\overline{Bo} < 0.4\overline{Md}+1.9$.
2. The Al addition significantly increases the yield strength of the alloy from 915 MPa (x = 0) to 1410 MPa (x = 1.0) at RT without obvious ductility loss due to the B2 type ordering. The Al1 alloy remain high yield strength up to 1088 MPa at 873 K. The greatly reduced density of the RHEAs as the result of Al addition also contributes to the dramatically increase of SYS. The combination of SYS and ductility of the $Ti_2ZrHf_{0.5}VNb_{0.5}Al_1$ RHEA is superior to most RHEAs from RT to 873 K.
3. The Al addition reduces oxidation weight gain of the $Ti_2ZrHf_{0.5}VNb_{0.5}Al_x$ RHEAs from the catastrophic oxidation of Al0 alloy after 24 h, to 112.1 mg·cm^{-2} of Al1 alloy after 50 h oxidation. The oxidation kinetics curves of Al0.5 and Al1 alloys followed the parabolic law before 24 h, but degraded into linear law after 24 h. The reason is attributed to the twisting and cracking of Al_2O_3 oxide layer and the formation of evaporable oxide V_2O_5 that accelerate the oxidation process.

In this paper, the process of transformation from single BCC structure to single B2 structure has been studied in depth. The results not only

provide experience for the design of RHEAs with B2 phase in the future, but also shed lights on the potential of structural application at elevated temperature for these RHEAs.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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